

Experiment 4: OFDMA Resource Block Allocation

Objectives, MATLAB Programs and Output Images

Objective 1

To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.

MATLAB Program

```
clc; clear; close all;

% System parameters
bandwidth = 10e6;           % 10 MHz
rb_size = 180e3;            % 180 kHz per RB

% Total Resource Blocks
total_RB = floor(bandwidth / rb_size);

% Users
users = 1:6;

% RB allocation (simple equal + remainder distribution)
base = floor(total_RB / length(users));

alloc = base * ones(1, length(users));
alloc(1:mod(total_RB, length(users))) = alloc(1:mod(total_RB, length(users))) + 1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')
disp(total_RB)

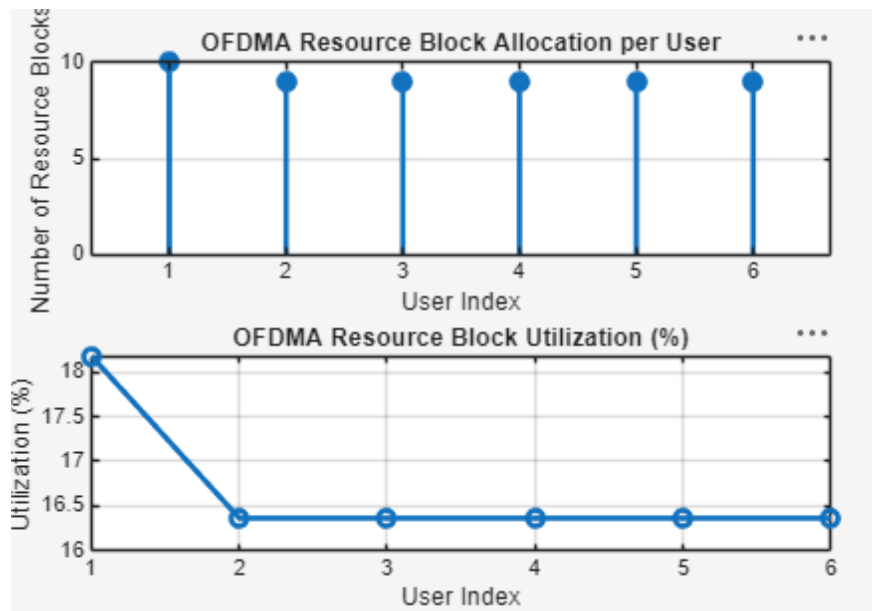
disp('RB Allocation per User:')
disp(alloc)

figure

% ----- Graph 1: RB Allocation per User -----
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: RB Utilization -----
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

Output Image – Objective 1



Objective 2

To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users.

MATLAB Program

```
clc; clear; close all;

% Total system resource blocks
total_RB = 50;

% Number of active users
users = 1:5;

% RB allocation to users (example scenario)
alloc = [12 10 9 8 11];

% Resource Block Utilization (%)
util = (alloc / total_RB) * 100;

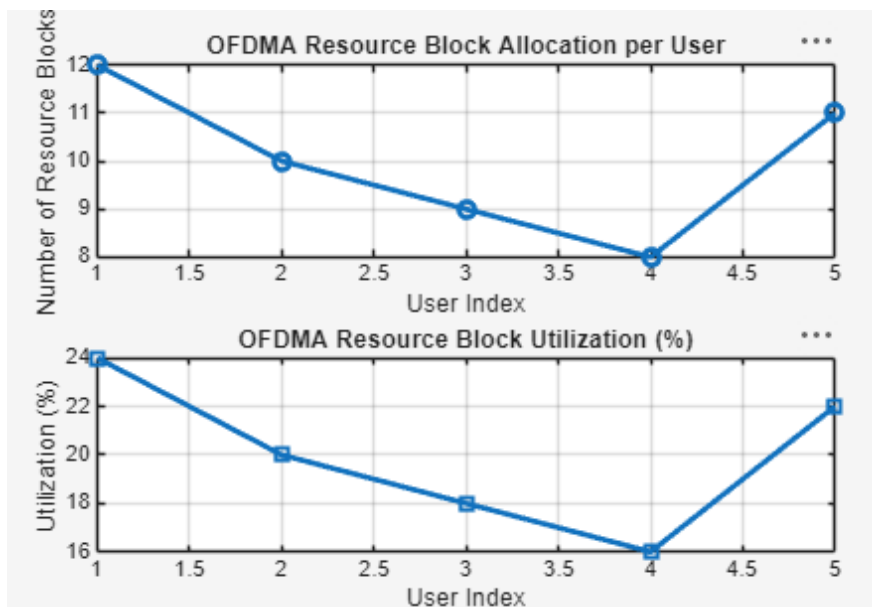
disp('Resource Block Utilization (%) per User:')
disp(util')

figure

% ----- Graph 1: RB Allocation -----
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on

% ----- Graph 2: Utilization (%) -----
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

Output Image – Objective 2



Objective 3

To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

MATLAB Program

```
clc; clear; close all;

% System parameters
rb_bw = 180e3;          % Bandwidth per RB (180 kHz)
users = 1:5;

% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];

% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

disp('User Throughput (Mbps):')
disp(throughput)

figure

% ----- Graph 1: Throughput per User -----
subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on

% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```

Output Image – Objective 3

